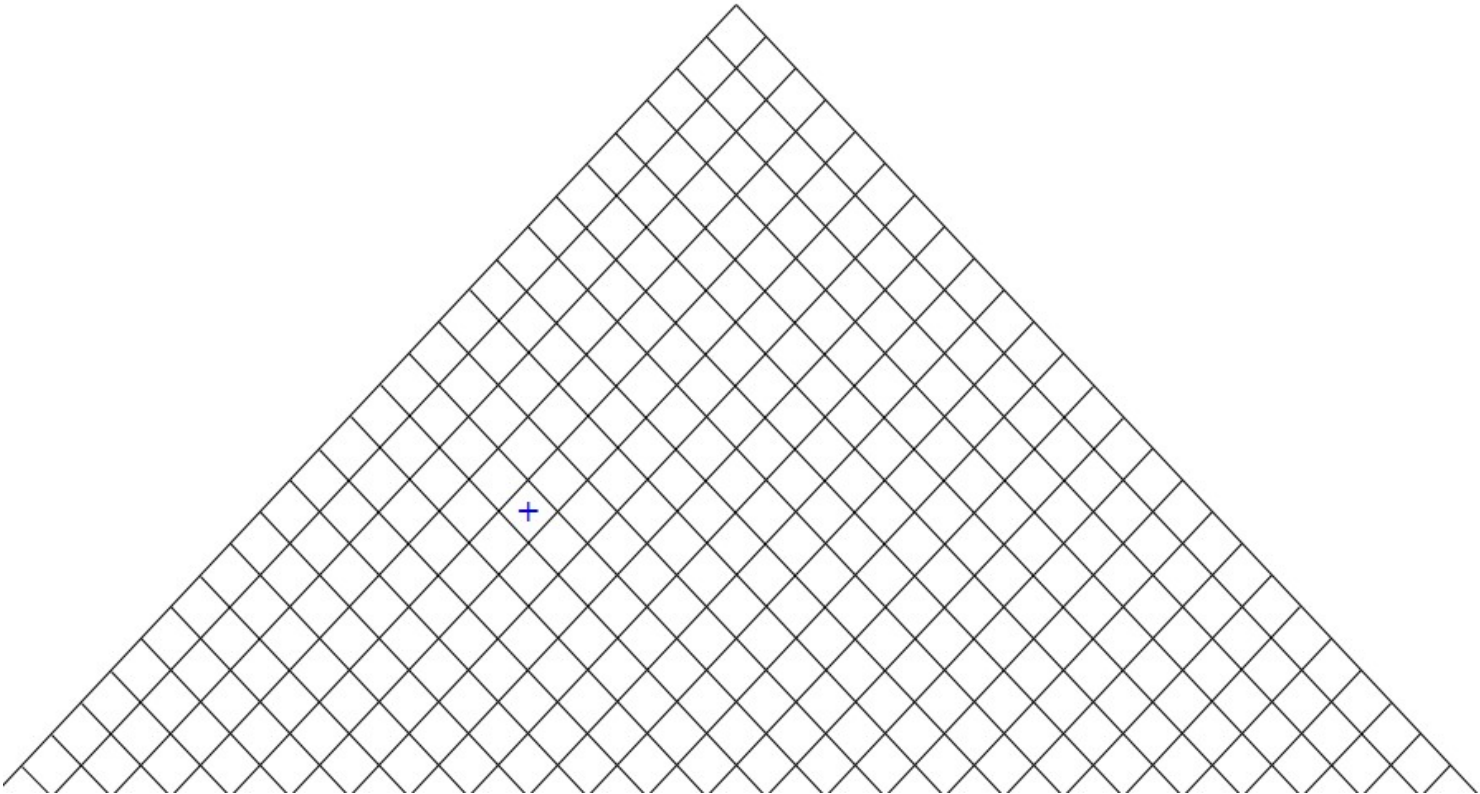


Title: QFD Level 1 -Smart water heater

Author: Sandip Patil

Date: 22-05-2026

Notes:



Legend		
		9
		3
		1
	Strong Positive Correlation	
	Positive Correlation	
	Negative Correlation	
	Strong Negative Correlation	
	Objective Is To Minimize	
	Objective Is To Maximize	
	Objective Is To Hit Target	

				Column #	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																		
				Direction of Improvement: Minimize (▼), Maximize (▲), or Target (x)	▼	▲	▲	▲		▲	▼	▲	▼	▲	▲		▲	▼		▼	▼	▼	▼	▼	▲	▲	▼	▲	▲	▲	Competitive Analysis (0=Worst, 5=Best)																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																
Row #	Max Relationship Value in Row	Relative Weight	Weight / Importance	Quality Characteristics (a.k.a. "Functional Requirements" or "Hows")	Anode Rod Sacrificial Rate (g/year)	Tank Coating Thickness (µm)	Scale Flush Valve Diameter (mm)	Anti-microbial Material Surface Area (%)	Max Pressure Resistance (MPa)	Voltage Surge Tolerance (V)	Thermal Expansion Coefficient (1/K)	First-Hour Delivery Rating (L/hr)	App/UI Thermal State-of-Charge Update Latency (ms)	Predictive Heating Algorithm Accuracy (%)	Thermostat Cut-off Accuracy (±°C)	Max Tap Outlet Temperature Limit (°C)	Thermal Mixing Valve Flow Rate (L/min)	Temperature Sensor Response Time (ms)	ELCB Trip Current (mA)	Leak Sensor Moisture Detection Threshold (mL)	Diagnostics Alert Latency (s)	Standing Heat Loss (kWh/24h)	Outlet Temperature Fluctuation (±°C/min)	Tank Stratification Efficiency (%)	Inlet Jetting Velocity (m/s)	Mean Time Between Failures (MTBF) (Hours)	App Health Diagnostics Coverage (%)	Predictive Maintenance Alert Lead Time (Days)	Warranty Supported Life (Years)																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																		</

